

Learning Objective 3.2: I can derive the array factor for an arbitrary linear array and apply pattern multiplication.

Show your work. A **Documentation** line at the end is required (write **None** if you did not collaborate).

1. **Beamwidth of the course array.** The ADALM-PHASER array has $N = 8$ patch elements on $d = 14$ mm centers and operates at 10.3 GHz. Use $\theta_{\text{HP}} \approx 0.886\lambda / (Nd \cos \theta_0)$ throughout.

- (a) Find the wavelength and the element spacing in wavelengths.

$$\lambda = \frac{3 \times 10^8}{10.3 \times 10^9} = 29.1 \text{ mm}$$

$$\frac{d}{\lambda} = \frac{14}{29.1} = 0.481$$

$$d/\lambda = 0.481$$

The aperture is $Nd = 112 \text{ mm} = 3.85\lambda$.

- (b) Compute the half-power beamwidth at broadside.

$$\begin{aligned} \theta_{\text{HP}} &= \frac{0.886}{3.85} = 0.230 \text{ rad} \\ &= 13.2^\circ \end{aligned}$$

$$\theta_{\text{HP}} = 13.2^\circ$$

- (c) Compute the half-power beamwidth with the beam steered to $\theta_0 = 45^\circ$.

$$\theta_{\text{HP}} = \frac{13.2^\circ}{\cos(45^\circ)} = \frac{13.2^\circ}{0.707} = 18.7^\circ$$

$$\theta_{\text{HP}} = 18.7^\circ$$

- (d) In one or two sentences, explain physically why the beam widens by $1/\cos \theta_0$ when the array is steered, even though no hardware changed.

A source in the direction θ_0 sees the aperture foreshortened: the projected length of the array normal to the beam is $Nd \cos \theta_0$, not Nd . Beamwidth is set by the aperture in wavelengths, so the effective aperture shrinks as the beam steers away from broadside and the beam widens by the same factor. At 60° the projected aperture is half its broadside value.

2. **Nulls and the first-null beamwidth.** Same array as Question 1: $N = 8$, $d/\lambda = 0.481$, so $Nd = 3.85\lambda$. The nulls of a uniform array are at $\sin \theta = \sin \theta_0 \pm m\lambda/(Nd)$ for $m = 1, 2, 3, \dots$, excluding multiples of N .

- (a) With the beam at broadside, find every null angle on the positive side of the pattern.

The tooth spacing in sine space is $\lambda/Nd = 1/3.85 = 0.260$.

nulls = 15.1, 31.3, 51.2°

$$\begin{aligned} m = 1 : \quad \sin(\theta) &= 0.260 \rightarrow 15.1^\circ \\ m = 2 : \quad \sin(\theta) &= 0.520 \rightarrow 31.3^\circ \\ m = 3 : \quad \sin(\theta) &= 0.780 \rightarrow 51.2^\circ \\ m = 4 : \quad \sin(\theta) &= 1.040 \rightarrow \text{no real angle} \end{aligned}$$

- (b) Compute the first-null beamwidth at broadside, and the ratio of it to the half-power beamwidth from Question 1(b).

FNBW = 30.1°

$$\text{FNBW} = 2 \arcsin\left(\frac{\lambda}{Nd}\right) = 2(15.1^\circ) = 30.1^\circ$$

$$\frac{\text{FNBW}}{\theta_{\text{HP}}} = \frac{30.1}{13.2} = 2.3$$

- (c) The array factor of an 8-element array has 7 nulls per period, but part (a) found only 3 on each side of broadside. Explain in one or two sentences where the others went.

The nulls are equally spaced in $\sin \theta$, and only those landing inside the visible region $|\sin \theta| \leq 1$ correspond to real directions. With $\lambda/Nd = 0.260$, the fourth null would require $\sin \theta = 1.04$, which is outside visible space. The pattern is periodic in sine space, but the antenna can only radiate into the visible part of it.

- (d) Steer the beam to $\theta_0 = 30^\circ$ and find the two nulls that bracket the main lobe, then the first-null beamwidth.

FNBW = 35.5°

$$\begin{aligned} \sin(\theta) &= 0.500 - 0.260 = 0.240 \rightarrow 13.9^\circ \\ \sin(\theta) &= 0.500 + 0.260 = 0.760 \rightarrow 49.4^\circ \\ \text{FNBW} &= 49.4^\circ - 13.9^\circ = 35.5^\circ \end{aligned}$$

The peak is 16.1° above the lower null and 19.4° below the upper one, so a steered beam is not symmetric about its own peak.

3. **Directivity of a uniform linear array.** Use $D \approx 2Nd/\lambda$ for a uniform broadside array of isotropic elements.

- (a) Compute the directivity of the 8-element course array, in linear units and in decibels.

$$D \approx 2(8)(0.481) = 7.7$$

$$D_{\text{dB}} = 10\log_{10}(7.7) = 8.9 \text{ dB}$$

$$D = 7.7 = 8.9 \text{ dB}$$

- (b) Suppose the same 8 elements were half-wave dipoles rather than patches. Estimate the total gain of the array in dBi.

Pattern multiplication becomes addition in decibels:

$$G_{\text{total}} = G_{\text{element}} + D_{\text{array}}$$

$$= 2.15 \text{ dBi} + 8.9 \text{ dB} = 11.1 \text{ dBi}$$

$$G_{\text{total}} = 11.1 \text{ dBi}$$

This holds while the element pattern is nearly flat across the main lobe.

- (c) Turning off the outer four elements leaves the center four running. State what happens to the directivity and to the half-power beamwidth, with numbers, and explain in one sentence why both changes have the same cause.

The aperture halves from 3.85λ to 1.92λ . Directivity halves, $D \approx 3.85 = 5.9 \text{ dB}$, a drop of exactly 3 dB, and the beamwidth doubles from 13.2° to about 27° . Both follow from the aperture in wavelengths: directivity is proportional to Nd/λ and beamwidth is inversely proportional to it, so any change in aperture moves them in opposite directions by the same factor.

4. **Sizing an array.** A ground-based X-band tracker at 10 GHz requires a half-power beamwidth of 2.0° and must steer to $\pm 60^\circ$.

- (a) Find the aperture length required at broadside, in millimeters and in wavelengths.

$$\lambda = 30 \text{ mm}, \quad \theta_{\text{HP}} = 2.0^\circ = 0.0349 \text{ rad}$$

$$Nd = \frac{0.886\lambda}{\theta_{\text{HP}}} = \frac{0.886(30)}{0.0349} = 762 \text{ mm}$$

$$\frac{Nd}{\lambda} = 25.4$$

$$Nd = 762 \text{ mm} = 25.4\lambda$$

- (b) The grating-lobe criterion is $d < \lambda / (1 + |\sin \theta_0|_{\text{max}})$. Find the largest usable spacing for this scan volume, then the element count if you build the array at $d = \lambda/2$.

$$d < \frac{\lambda}{1 + \sin(60^\circ)} = \frac{\lambda}{1.866} = 0.536\lambda = 16.1 \text{ mm}$$

$$N = \frac{762}{15} = 50.8 \rightarrow 51 \text{ elements}$$

$$d_{\text{max}} = 0.536\lambda$$

$$N = 51$$

Round up, never down: rounding down misses the beamwidth specification.

- (c) A colleague proposes $d = 0.9\lambda$ instead, which reaches the same 25.4λ aperture with 28 elements rather than 51 – a large saving in phase shifters and amplifiers. Evaluate the proposal: compute where the grating lobe lands at the edge of the scan volume, and state whether you would accept it.

Grating lobes appear at $\sin \theta_g = \sin \theta_0 \pm \lambda/d$. With $d = 0.9\lambda$, $\lambda/d = 1.111$. At broadside, $\sin \theta_g = 1.111$ is outside visible space and nothing is wrong. Steered to 60° :

$$\sin(\theta_g) = 0.866 - 1.111 = -0.245 \rightarrow -14.2^\circ$$

A full-strength second beam appears at -14.2° whenever the array looks at the edge of its volume, so the tracker would report targets in the wrong direction. Reject the proposal: the saving in elements is real, but the array no longer meets the scan requirement.

- (d) The beamwidth requirement must hold across the whole scan volume, not just at broadside. State the aperture this actually demands and the resulting element count at $d = \lambda/2$.

The beam is widest at the edge of the volume, so size for $\theta_0 = 60^\circ$:

$$Nd = \frac{762}{\cos(60^\circ)} = 1524 \text{ mm} = 50.8\lambda, \quad N = \frac{1524}{15} = 102$$

The last 60° of scan doubles the array. Whether that is worth paying for is a system question, and it has to be asked before the aperture is fixed.

5. **Reading a measured sweep.** A student sweeps the course array at 10.3 GHz ($\lambda = 29.1$ mm) with the GUI and records a main lobe 13.1° wide at half power, a null-to-null width of 28.5° , and a first sidelobe at -11.5 dBc.

- (a) From the measured half-power width alone, infer the aperture in millimeters and the number of active elements on 14 mm centers.

$N =$

8

$$Nd = \frac{0.886\lambda}{\theta_{HP}} = \frac{0.886(29.1)}{0.2287} = 113 \text{ mm}$$

$$N = \frac{113}{14} = 8.1 \rightarrow 8 \text{ elements}$$

- (b) Theory puts the first-null beamwidth at 30.1° and the student measured 28.5° . The sweep steps the commanded angle by the phase-shifter resolution of 2.8125° . Explain the discrepancy in one or two sentences.

The sweep samples the pattern every 2.8125° , so the recorded trace almost never lands on the exact null angle of 15.1° . Nulls are narrow, sharp features, so the apparent null position moves to the nearest grid point and the measured null-to-null width reads one to three degrees off. The half-power points sit on a gentle slope and interpolate far more reliably, which is why beamwidth is quoted at half power.

- (c) The measured first sidelobe is -11.5 dBc, while theory for a uniform 8-element array gives -12.8 dBc. Is this evidence of a fault? Answer in one or two sentences.

No. A uniform array's first sidelobe is expected between -11 and -13 dBc on this hardware once element-to-element amplitude and phase errors, the coarse sweep grid, and the finite noise floor are included, and -11.5 dBc sits inside that band. A sidelobe several decibels high, or a main lobe that is not at the commanded angle, would be evidence of a fault.

- (d) A second sweep, taken after a lab partner changed the element gains, shows a main lobe 27° wide with no visible sidelobes above the noise floor. State which aperture configuration this is and what it implies about the number of active elements.

$N =$

4

$$\theta_{HP} = 27^\circ = 0.471 \text{ rad}$$

$$Nd = \frac{0.886(29.1)}{0.471} = 54.7 \text{ mm} \approx 1.9\lambda$$

That is four elements at 14 mm spacing, so the center four are on and the outer four are off. With only four elements the sidelobes rise to about -11 dBc, so "none visible" points instead to a tapered set of element gains rather than a clean four-element aperture.

Documentation: _____